

Introduction and Related Work (4/5)

- There are different ways to linearize a high dimensional system that is stochastic in nature.
- There are multiple Ising-inspired models that were developed to compare simplification strategies.
- These models look at how the simplifications distort the overall behavior.
- The introduction currently feels slightly repetitive, as both the first and second paragraphs introduce the Ising-inspired models. To improve the narrative flow and create a stronger hook, I recommend restructuring the section into a format:
 - **Paragraph 1:** Focus entirely on introducing the broader problem (the challenge of linearizing stochastic, high-dimensional non-linear systems).
 - **Paragraph 2:** Introduce the Ising model specifically as the ideal testbed for evaluating these high-dimensional systems.
 - **Paragraph 3:** Keep as is.
 - **Paragraph 4:** Move the specific model descriptions from your original first paragraph down to here.
- Background is well thought out where the Ising model is clearly described and the for simplifications.

Problem Statement (4/5)

- The problem statement is scattered throughout the whole report where pieces can be found in the Introduction and Background sections.
- This paper would benefit from a dedicated problem statement where it is clarified and accompanied by a hypothesis.
- This would enhance the well thought out method section of System Description, Model Simplifications and Experimental Protocol.

Results (4/5)

- Figure 1 is a great visual representation of the 5 tested models which indicates mean magnetism over time with confidence intervals
- From the results section model 1 was the reference dynamics and the transient restructuring/simplification models (model 5) visually mirrored model 1 while the others had a smoother transition to equilibria.

- The author suggest that the parameters J,h,T, together with time, appear sufficient to describe large-scale model behavior across different simplifications.
- Instead of presenting a scattered grid of combinations, organize the results systematically by isolating variables. Hold two parameters constant while sweeping the third (e.g., hold J and h constant while varying T) so the reader can clearly evaluate the marginal effect of each parameter on model divergence.
- It would be beneficial to add a comparison metric to baseline model 1 such as root mean squared errors. Additionally, convergence time figures will strengthen your narrative that models 3 and 4 had slower convergence rates.

Conclusion (5/5)

- Great summary of the work done and what the main contribution of the paper is.
- Future work is a great addition.

Professionalism (4/5)

- The paper is formatted nicely in a standard two-column layout and the writing is technically sound and well-organized.
- In the Introduction there is an “a” in italics which is inconsistent with the rest of the text.
- Missing several in-text references especially in the background and introduction section.

Overall (4.2/5)